Amit Kumar
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LEADERSHIP EXPERIENCE

Senior Technical Lead with 12+ years of experience in embedded systems, networking, and Linux-based platforms. Specialized in C programming, OpenWrt firmware development, and router platform engineering for global clients including Cisco, NEC, and Juniper. Delivered high-perform solutions, system stability & implement advanced features for Wi-Fi standards (802.11 a/b/g/n/ac/ax).

EDUCATION

2011 Roland Institute of Technology B.Tech (Computer Science) 7/10

EXPERIENCE

PROFESSIONAL EXPERIENCE: Having **12+ years** of experience in software development in **C**, **Linux & VxWorks** Platform.

- **HCLTech**(Noida) 4 years
  Duration From December 2021 till date
  Designation Senior Tech Lead
  Department Asr9k routers platform, NEC VEOS
  Technology C, Linux, networking (UDP/TCP), RHEL10 RPM porting, Agile
- **Inventum** (Noida) 2 Years
  Duration From December 2019 to Nov 2021
  Designation Lead Engineer
  Department Research & Development
  Technology C, Linux, networking (UDP/TCP), OpenWrt, CPE, Agile
- **Incedo Inc** (Gurgaon) 1 Year 8 months
  Duration From October 2017 to June 2019
  Designation Senior Software Developer
  Department Research & Development
  Technology C, Linux, networking (UDP/TCP), VxWorks, Agile
- **Aricent** (Gurgaon) 2 Years 6 Months
  Duration From December 2014 to May 2017
  Designation Senior Software Developer
  Department Research & Development
  Technology C and Linux
- **Towards Vision Technologies Pvt. Ltd.** (Gurgaon) 2 Years 8 Months
  Duration From April 2012 to December 2014
  Designation Software Developer
  Department Research & Development
  Technology C and Linux

SKILLS

Programming Languages : C, Shell-Script, Python-Basics
Repository Tools & Kits : GIT**,** SDLC, Agile
Operating Systems : Linux, Embedded Systems, VxWorks, Cisco-IOS, OpenWrt
Protocols : Wi-Fi (802.11 a/b/g/n/ac/ax), All Networking Protocols
Datacomm/Telecomm : Asterisk (IPBX), Router Firmware, OpenWrt, Qualcomm Atheros
Linux Packaging : .deb(Ubuntu, Mint) & .rpm(RHEL, CentOS, Fedora)
Tools : tcpdump/Wireshark, nping/trex/iperf2, ifconfig/ping/traceroute/nslookup

PROJECTS

Project 1:

Client: **Cisco & NEC**	Environment: HCLTech
Project Name: **Asr9k routers, NEC VEOS**	Duration: December2021- till date

- **Cisco Asr9k routers for Cisco:**
  - Bug fixing, maintenance project, CBC module owner, CBC fw versions patch, client interaction & support, MPLS Basics.
  - IPC socket/select/poll, network troubleshoot, Customer issue & support
  - Used troubleshoot tools like tcpdump/wireshark, nping/trex/iperf2, ifconfig/ping/traceroute/nslookup
    - **Skill Used** C ,TCP-socket , mysql, VMware, networking (UDP/TCP), multithreading, shell scripting
- **NEC VeOs for NEC Japan:**
  - porting work from RHEL8 to RHEL10 for Linux packages supporting VEOS platform
  - Customer issue & support
  - Automation development
  - Linux process and memory management
  - IPC networking sockets (UDP/TCP)

- **Skill Used** Porting to RHEL10 RPM, VMware, feature support for procps commands, bugfix
- **Technology Used** C, Linux, GIT, complex cpu & memory %age calculations, router platform, shell scripting
- **Package Porting** procps, time, free, top, vmstat, CI/CD, pgrep, kill, pmap, ps, strace

Project 2:

Client: **Inventum**	Environment: **Inventun R&D**
Project Name: **Udaya-AP, Small Business Router**	Duration: December2019- Dec 2021

- Openwrt build image and porting stuff for Qualcomm SDK & Kenstel SDK
- Platform setup for openwrt-19.07.7 -ipq40xx
- enable 16(3/13) SSID support for AP for wifi standards 802.11b/a/g/n/ac, ax-basics
- Features – 1. enable reverse-ssh features, 2. Developed DHCP relay from scratch, 3. enable mqtt json req-reponse setup for remote monitoring Aps, 4. Enable cascading of Aps in bridge mode (AP1—AP2—AP3 ..)
- Openwrt uci support added for mac learn limit setting, enable list view for connected client for AP
- Develop parental control (using uci & /etc/init.d/firewall via scipt)
- Bug fixing & field Customer issues
  - **Skill Used** : feature support, bugfix
  - **Tools & Kits** : tcpdump, Wireshark, strace, perf
  - **Technology Used** : C, Linux, openwrt, UCI, LUCI web, 802.11b/a/g/n/ac/ax, TCP/IP

Project 3:

Client: **Belden, Hirschmann**	Environment: **Incedo R&D**
Project Name: **DX, Rx , HiOs(Hirschmann)**	Duration: Oct 2017 – June 2019

- Feature Enhancement of "Persistent Tracing Mechanism for Belden DX-940 Router.
- Worked on DHCP option 82
- Feature Enhancement of "Implement Encryption for config file/logfile/Build Image Snapshot" while transfer to remote machine
- Associated with the development of Hirschmann's HIOS product for following activities
  1. Set of networking routers & switches required to be ported from VxWorks to Linux.
  2. Creation of Compilation Environment in Linux.
  3. Requirement gathering from client. Bug Fixing, Debugging in VxWorks Shell.
  4. Ethernet IP Development for HIOS linux based devices.
  5. USBC gadget implementation over usbc port.
  6. GARP/GVRP protocol bux fixing.
  - **Skill Used** Feature Enhancement, Bug Fixing, Unit and Functional Testing
  - **Technology Used** C, Linux, VxWorks, OpenWrt, Task Health Monitoring

Project 4:

Client: **Juniper Networks**	Environment: **Aricent**
Project Name: **CTP, VCTP**	Duration: Dec 2014 – May 2017

Juniper Networks CTP Series Circuit to Packet Platforms provide the advanced technology and features required to reliably transport circuit-based applications across IP/MPLS networks. CESoPSN and SAToP bundle tunneling for serial transmission. Involved in a JITC.

- **Skill Used:** Bug fixing. Virsh-virtualization for CTP.
- **Technology Used:** C, Linux, Nova Drivers, SNMP, kernel-3.10 migration, virsh

Project 5:

Client: Czentrix	Environment: **Czentrix**
Project Name: **Asterisk Telephony**	Duration: 4 Months

- **FTP client :** Developed an FTP client to transfer recording files between servers while adhering to specific predefined rules.
- **TvtShell** : Customized shell to provide access to own servers on customer's interest. The whole idea and coding done in C on Linux environment.
- **HddClone :** Dual hard disk setup for a server can autonomously reboot & shift to backup HDD on crash or hard disk failure
- **Health check system :** Developed health monitoring system to assess the status of multiple hard drives connected to a server.
- **Agent Interface Server :** An interface has been designed to route all telephony events arriving through ports 5039 and 7001 to ports 5139 and 8001, enabling the UI to process them. Additionally, a message mediator has been implemented to ensure event transmission continues even if the telephony engine (T.E.) is down.

  - **Skill Used** **:** Feature Enhancement, Bug Fixing, Unit and Functional Testing
  - **Tools & Kits** **:** tcpdump, Wireshark, strace, perf**. Editline**, crone-tab, system calls
  - **Technology Used** **:** C**, TCP-socket , mysql, multithreading,** feature support, bugfix